Lecture 8

Tuesday, 8 September 2026 7:19 PM

Random Variables / CDF

- Discrete → PMF
- Conts → PDF

EXPECTATION OF RV

RV gives a real value to every experiment.

EXPECTATION → 'mimic' avg of the values one expects to see if we draw multiples samples from the RV.

X discrete RV. S_X → support

N samples from P_X → $k \in S_X$, N_k observations

Avg of these values → $\dfrac{\sum_{k \in S_X} k \cdot N_k}{N} = \sum_{k \in S_X} k \cdot \dfrac{N_k}{N}$ ✓

$\dfrac{N_k}{N} \to P_X(k)$

$P_X(k) \to P(X = k)$

$$\Rightarrow E(X) \triangleq \sum_{k \in S_X} k \cdot P_X(k)$$

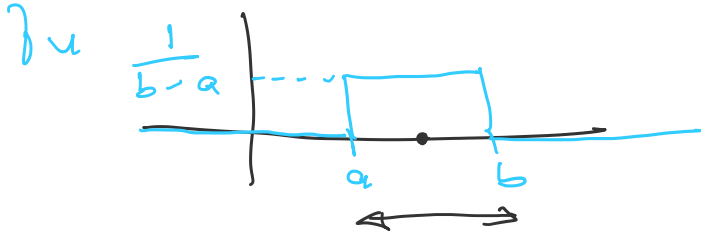


For a conts RV $X \sim f_X(\cdot)$,

$$E(X) = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx$$

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Eg $U \sim \text{uniform}[a, b]$



$$E[U] = \int_{-\infty}^{\infty} x f_X(x)\,dx$$

$$= \int_a^b x \cdot f_X(x)\,dx$$

$$= \frac{1}{b-a} \int_a^b x \cdot dx$$

$$= \frac{1}{b-a} \cdot \left[\frac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2\cdot(b-a)} = \boxed{\frac{a+b}{2}}$$

$(b-a)(b+a)$

Eg $X \sim Ber(p)$ $\quad P(X=1) = p$, $\quad P(X=0) = 1-p$

$$\boxed{E(X) = p} = \sum_{k \in S_X} k \cdot P_X(k) = \underbrace{1 \cdot P(X=1)}_{p} + \underbrace{0 \cdot P(X=0)}_{(1-p)}$$

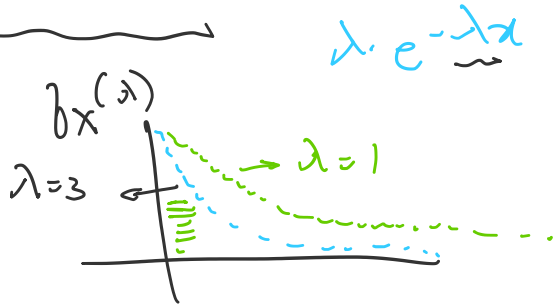
Eg $X \sim Exp(\lambda)$, $\lambda > 0$

$$f_X(x) = \lambda \cdot e^{-\lambda x}, \quad x \geq 0$$

$$E(X) = \int_0^{\infty} x \cdot \lambda \cdot e^{-\lambda x}\,dx$$

$$= \lambda \cdot \int_0^{\infty} x \cdot e^{-\lambda x}\,dx$$

$$= \frac{1}{\lambda}$$



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Eg $N \sim Bin(n, p)$ $S_N = \{0, 1, 2, \ldots n\}$

$$E(N) = \sum_{k=0}^{n} k \cdot P_X(k)$$

$$= \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k (1-p)^{n-k} = \boxed{n \cdot p}$$

Eg $X \sim N(\mu, \sigma^2)$



$E(X) = \mu$

Eg $\boxed{X \sim Geo(p)}$ $\boxed{P_X(k) = p(1-p)^{k-1}}$ $k \in \{1, 2, 3, \ldots\}$

$$E(X) = \sum_{k=1}^{\infty} k \cdot p(1-p)^{k-1}$$

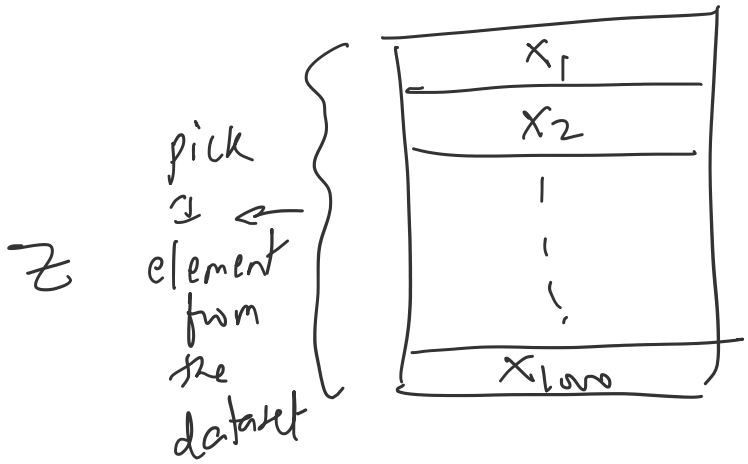
$$\| \quad \sum_{k \in S_X} k \cdot P_X(k) = \frac{1}{p}$$

$E(X) \approx \frac{\sum_{i=1}^{1000} X_i}{1000}$

Exp1 → Toss 1 → T, Toss 2 → T, Toss 3 → H

$X_1 = 3$

Exp2 → T T T T H, $X_2 = 5$



$$E(Z) = \frac{X_1 + X_2 \cdots + X_{1000}}{1000}$$

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$Z \in \{X_1, X_2, X_3 \ldots X_{100}\}$

$\downarrow \frac{1}{100} \quad \downarrow \frac{1}{100} \quad \downarrow \frac{1}{100}$

Eg E-Retailer → Decide how much of a product to stock → Q

Demand D is uncertain

$D \sim \text{Unif.}[0, 100]$

For each product, Cost → Rs 60

Sale price → Rs 100

Revenue on the day.

↳ $100 \cdot \min(D, Q)$

Cost → $60 \cdot Q$

* $\text{Profit} = 100 \cdot \min(D, Q) - 60 \cdot Q$

* $E[\text{Profit}] = E[100 \cdot \min(D, Q) - 60 \cdot Q]$

$g(D, Q)$

$$= \int_0^{100} g(x, Q) \cdot f_D(x) \cdot dx$$

$g(x, Q)$ for $x \in [0, 100]$

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$$= \frac{1}{100} \int_0^{100} g(x, Q) \cdot dx$$

$$= \frac{1}{100} \left[\int_0^{Q} g(x, Q)\, dx + \int_{Q}^{100} g(x, Q)\, dx \right]$$

$$\int_0^{Q} g(x, Q)\, dx = \int_0^{Q} [100 \cdot x - 60 Q]\, dx$$

$$= 100 \left[\frac{x^2}{2} \right]_0^{Q} - 60Q \int_0^{Q} 1 \cdot dx$$

$$= 100 \cdot \frac{Q^2}{2} - 60 Q^2 = -10 Q^2$$

$$\int_{Q}^{100} g(x, Q)\, dx = \int_{Q}^{100} (100 Q - 60 Q)\, dx$$

$$= 40Q \cdot \int_{Q}^{100} 1 \cdot dx$$

$$= 40Q (100 - Q)$$

$$E[\text{Profit}]$$

$$= \frac{1}{100} \left[-10 Q^2 + 4000 Q - 40 Q^2 \right]$$

$$= \boxed{40Q - .5 Q^2}$$



$$\frac{d}{dQ} E(\text{Profit}) = 40 - 2 \times .5 \times Q$$

$$\Rightarrow 40 - Q = 0$$

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$\Rightarrow \boxed{Q^* = 40}$ ✓

$g(D, Q) = 100 \cdot \min(D, Q) - 60Q$

$D \leq Q \rightarrow 100D - 60Q$ ✓ *

$\rightarrow D \geq Q \rightarrow 100Q - 60Q = 40Q$ ✓ *

$E[D] = \frac{100+0}{2}$ $\quad E[g(D, Q)]$

X RV , $g(x)$ $\quad E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$

$D > Q$

$D = 30$ kg $\quad Q = 10$ kg.

Revenue $\rightarrow 100 \times 10$

Cost $\rightarrow 60 \times 10$

Profit $\rightarrow 40 \times 10$

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